

AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	41.0 / Female
Department	General Medicine
Diagnosis	Urethritis
UTI Classification	Recurrent UTI
Sample Type	Urine

Clinical Presentation

Chief Complaints: Burning urination;Urgency

Comorbidities: None

Surgical History: None

Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	37.0	%	20 - 40
Wbc	9100.0	cells/ μ L	4000 - 11000
Polymorphs	64.0	%	40 - 75
Crp	9.0	mg/L	< 10
Rft Serum Creatinine	0.7	mg/dL	0.6 - 1.3
Serum Uric Acid	4.2	mg/dL	3.5 - 7.2
Blood Urea	26.0	mg/dL	7 - 20
Cue Pus Cells	12.0	/HPF	0 - 5
Epithelial Cells	3.0	/HPF	0 - 5
Proteins	Trace		Negative
Rbc	1.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	Streptococcus agalactiae
Bacteria Type	Gram-positive coccus (Group B Streptococcus; significant in pregnancy)

Antibiotic Susceptibility Profile

Predicted Sensitive	Daptomycin
Predicted Resistant	Azithromycin

Recommended Therapy

Daptomycin

Dosage: 6 mg/kg IV once daily (e.g., 400 mg for a 70 kg adult) infused over 30 minutes

Precautions: Normal renal function (serum creatinine 0.7 mg/dL) allows standard dosing. Monitor serum creatine kinase weekly and discontinue if CK >5× upper limit or patient develops myalgia. Avoid use if severe renal impairment or in patients with known hypersensitivity to daptomycin.

Rationale: Daptomycin is active against Gram-positive cocci, including Streptococcus agalactiae, and is listed as sensitive in the prediction. It provides reliable bactericidal activity for serious GBS infections and is the only antibiotic in the sensitive list suitable for this patient.

Antibiotic Drug History

Clinical Summary

Infection & Microbiology

- Lower-tract, recurrent urethritis caused by Streptococcus agalactiae (Group B Strep).
- Predicted resistance to azithromycin; sensitivity confirmed for daptomycin.

Antibiotic Recommendation

- Daptomycin 6 mg/kg IV once daily (≈400 mg for a 70 kg adult) over 30 min.
- Rationale: potent, bactericidal activity against Gram-positive cocci, specifically GBS, and the only drug in the sensitive list appropriate for this patient's profile.

Key Precautions

- Normal creatinine (0.7 mg/dL) permits standard dosing.
- Monitor serum creatine kinase weekly; discontinue if CK > 5× ULN or if myalgia develops.

- Avoid in severe renal impairment or in patients with known hypersensitivity to daptomycin.

This regimen addresses the resistant organism while minimizing potential toxicity, ensuring effective treatment of the recurrent urethritis.